

Effects Of Macroeconomic Indicators On Indian Stock Returns: A Deep Learning Perspective

A Project Submitted in Partial Fulfillment
of the Requirements for the Degree of
Master of Science in Data Science

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Academic Year: 2025–2026

Certificate

This is to certify that the project report entitled *Effects Of Macroeconomic Indicators On Indian Stock Returns: A Deep Learning Perspective*, carried out at the Department of Data Science, The Institute of Science, Mumbai, in partial fulfillment for the award of the degree of Master of Science in Data Science, Dr. Homi Bhabha State University, Mumbai, is a record of bonafide work carried out by **Mr. Rohan Kashinath Shinde (Seat No. DS2405)**, **Mr. Suraj Santosh Mishra (Seat No. DS2411)**, under the supervision and guidance of **Ms. Juilie Shinde** during the academic year 2025 – 2026.

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Declaration

I hereby declare that the project work entitled “*Effects Of Macroeconomic Indicators On Indian Stock Returns: A Deep Learning Perspective*” carried out at the Department of Data Science, The Institute of Science, Mumbai, is a record of an original work done by me under the guidance of **Ms. Juilie Shinde**, The Institute of Science, and this project work is submitted in the partial fulfillment of the requirements for the award of the degree of Master of Science in Data Science, Dr. Homi Bhabha State University, Mumbai. The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.

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Acknowledgement

We would like to express our sincere gratitude to Dr. Selby Jose, Head of Department, The Institute of Science, Mumbai, for his valuable supervision, constructive guidance and encouragement throughout the course of this research project. His expertise in data science and financial modelling significantly shaped the direction and quality of this work. We extend our heartfelt thanks to all faculty members of the Department of Data Science for their academic support and motivation during our postgraduate studies. We are also grateful to The Institute of Science, Mumbai, and Dr. Homi Bhabha State University for providing the necessary academic resources, computing infrastructure, and an intellectually stimulating research environment. Our appreciation extends to the Reserve Bank of India (RBI), the Ministry of Statistics and Programme Implementation (MOSPI), and Yahoo Finance for making publicly available macroeconomic and financial data, which formed the backbone of this study.

We also acknowledge the open-source community behind Python libraries including TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Statsmodels, XGBoost, and keras-ten, without which this work would not have been computationally feasible.

Finally, we thank our families and friends for their constant moral support and understanding during the demanding phases of this project.

Rohan Shinde Suraj Mishra

Abstract

Macroeconomic indicators are widely recognized as significant drivers of stock market performance, particularly in emerging economies such as India, where financial markets are closely intertwined with broader economic conditions. This study examines the relationship between key macroeconomic variables - Gross Domestic Product (GDP), inflation, repo rate, INR/USD exchange rate, and Foreign Institutional Investment (FII) - and the performance of the Indian stock market, represented by the NIFTY 50 and SENSEX indices.

A dataset covering 2004 to 2025, comprising 87 quarterly observations, was collected from the Reserve Bank of India (RBI), the Ministry of Statistics and Programme Implementation (MOSPI), and Yahoo Finance. Missing values were handled using forward-fill interpolation, and Min-Max normalization ensured consistent feature scaling. Temporal sequences of four quarters were constructed to capture lagged macroeconomic dependencies.

Eight forecasting models were developed and evaluated: ARIMA and VAR (econometric), Random Forest, XGBoost, and Support Vector Machine (machine learning), and LSTM, GRU, and Temporal Convolutional Networks (deep learning). Model performance was assessed using RMSE, R^2 , MAPE, directional accuracy, and AUC, with five-fold Time series split cross-validation applied due to limited data.

Results show deep learning models outperform traditional approaches. The GRU model achieved the best performance with the lowest RMSE (2,255,120), highest R^2 (0.6597), lowest MAPE (4.69%), and highest directional accuracy (75.00%). LSTM also performed well, while ARIMA, VAR, and ensemble models showed higher errors. These findings highlight the strong influence of macroeconomic factors on stock market behavior and demonstrate the effectiveness of GRU in capturing nonlinear temporal patterns.

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Chapter 1

Introduction

1.1 Background

Financial markets serve as one of the most sensitive barometers of an economy's health. In India, the two primary stock market indices, NIFTY 50 and SENSEX, aggregate the performance of the country's largest publicly listed companies and are closely monitored by investors, policymakers, and analysts alike. These indices do not operate in isolation; they respond dynamically to shifts in macroeconomic fundamentals, both domestic and global.

The relationship between macroeconomic factors and stock market performance has been the subject of extensive academic inquiry since Fama's seminal work on efficient markets in 1970. Subsequent studies by Chen, Roll, and Ross (1986) established theoretical and empirical linkages between macroeconomic forces such as inflation, interest rates, and industrial output, and asset prices. In the Indian context, these relationships are particularly salient, as the economy has undergone significant structural transformation since the liberalization of the early 1990s, with capital markets increasingly integrated with global financial systems.

Traditional econometric models, including Autoregressive Integrated Moving Average (ARIMA) and Vector Autoregression (VAR), have long served as standard tools for financial forecasting. However, these models are constrained by assumptions of linearity and stationarity that may not adequately capture the complex, nonlinear dynamics inherent in financial time series. The rapid advancement of machine learning and deep learning methodologies in recent years has opened new avenues for modeling such complexities. Recurrent neural network architectures, especially Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, have demonstrated remarkable capability in learning sequential patterns in data, making them particularly well-suited for time series forecasting tasks.

1.2 Problem Statement

Despite a growing body of literature on machine learning applications in finance, comparative evaluations of traditional econometric methods against state of the art deep learning architectures for predicting macroeconomic driven stock market trends in the Indian context remain limited. Furthermore, with the increasing availability of multi source macroeconomic data, there is a need for models capable of harnessing multivariate temporal information to generate reliable forecasts.

This study addresses the following research problem: Given quarterly macroeconomic indicators spanning two decades, can deep learning models significantly outperform traditional econometric and machine learning approaches in predicting Indian GDP trends and stock market behaviour as reflected by NIFTY and SENSEX indices?

1.3 Objectives

The specific objectives of this study are:

- To analyze the relationship between key macroeconomic variables (GDP, inflation, repo rate, INR/USD exchange rate, and FII) and Indian stock market performance (NIFTY 50 and SENSEX).
- To develop and evaluate forecasting models using traditional econometric approaches (ARIMA and VAR), classical machine learning algorithms (Random Forest, XG-Boost, and SVM), and deep learning architectures (LSTM, GRU, and TCN).
- To compare model performance using multiple evaluation metrics: RMSE, R^2 , MAPE, directional accuracy, and AUC.
- To identify the optimal predictive model for Indian macroeconomic and stock market forecasting.
- To provide practical insights for investors, financial analysts, and policymakers.

1.4 Scope of the Study

This study is confined to quarterly macroeconomic data for the Indian economy covering the period Q2 2004 to Q1 2025, yielding 87 observations. The stock market variables of interest are NIFTY 50 High and SENSEX High prices. The macroeconomic predictors include GDP (at current prices), RBI repo rate, CPI/WPI inflation, INR/USD exchange rate, and FII equity, debt, and total flows. The target variable for deep learning prediction is India's GDP, which is used as a proxy for macroeconomic health. The study does not

incorporate intra-day or monthly data, nor does it account for geopolitical shocks, global commodity prices, or investor sentiment indices, which are acknowledged as areas for future research.

1.5 Organization of the Report

This report is organized into five chapters. Chapter 1 introduces the research context, problem statement, objectives, and scope. Chapter 2 reviews existing literature on macroeconomic-stock market linkages and machine learning applications in finance. Chapter 3 details the dataset, preprocessing pipeline, feature engineering, and the eight modelling approaches employed. Chapter 4 presents comprehensive experimental results and comparative analysis across all models. Chapter 5 concludes the study with key findings, limitations, and directions for future research.

Chapter 2

Literature Review

2.1 Overview of Existing Work

2.1.1 Macroeconomic Drivers of Stock Market Performance

The theoretical foundation for macroeconomic-financial market linkages was established by Fama (1970), whose efficient market hypothesis posited that stock prices fully reflect all available information, including macroeconomic fundamentals. Chen, Roll, and Ross (1986) empirically demonstrated that macroeconomic forces such as industrial production, inflation, term structure of interest rates, and default risk premiums systematically explain stock return variation. This arbitrage pricing theory (APT) framework has since guided extensive empirical research across global markets.

In the Indian context, several studies have investigated the influence of macroeconomic variables on NIFTY and SENSEX returns. Research by Bhattacharya and Mukherjee (2003) and Naik (2013) found significant relationships between exchange rates, FII flows, and stock prices using Granger causality and cointegration tests. Studies by Patel (2012) and Pal and Mittal (2011) employed regression-based frameworks to establish that inflation and interest rates are key determinants of Indian equity market performance.

2.1.2 Traditional Time Series Forecasting

ARIMA models, introduced by Box and Jenkins (1976), have long been the workhorse of univariate time series forecasting in economics and finance. Their strength lies in capturing autocorrelation structures in stationary series. VAR models extend ARIMA to multivariate settings by capturing linear inter-dependencies across multiple time series simultaneously. While robust for short-horizon forecasts under linear assumptions, both ARIMA and VAR have well-documented limitations when applied to nonlinear financial time series with structural breaks and regime changes.

2.1.3 Machine Learning Approaches in Financial Forecasting

Patel et al. (2015) demonstrated that ensemble machine learning methods, particularly Random Forest and Support Vector Machines, outperform traditional regression models in stock index prediction tasks. XGBoost, introduced by Chen and Guestrin (2016), has since become a competitive benchmark for tabular financial data due to its regularization capabilities and gradient boosting framework. However, these methods treat each sample independently and require manual feature engineering to capture temporal dependencies.

2.1.4 Deep Learning for Financial Time Series

The application of recurrent neural networks to financial forecasting was pioneered by Fischer and Krauss (2018), who demonstrated that LSTM networks significantly outperform random walk benchmarks and classical statistical models for S&P 500 constituent returns. Gu, Kelly, and Xiu (2020) provided a comprehensive evaluation of machine learning methods in empirical asset pricing, confirming the superiority of deep learning architectures for capturing nonlinear factor interactions.

Gated Recurrent Units (GRUs), proposed by Cho et al. (2014), offer a computationally efficient alternative to LSTMs by using two gates (reset and update) instead of three (input, forget, output), often achieving comparable or superior performance on financial prediction tasks with limited data. Temporal Convolutional Networks (TCNs), introduced by Bai et al. (2018), employ dilated causal convolutions to achieve long-range temporal dependencies without the vanishing gradient issues of RNNs.

2.2 Critical Analysis

A critical review of existing work reveals several important themes. First, most studies focus on either stock return prediction or macroeconomic forecasting independently, without jointly modeling the macroeconomic financial market interface. Second, the majority of deep learning applications in Indian financial markets rely on high-frequency price data (daily or intra-day) rather than macroeconomic fundamentals at quarterly granularity. Third, comparative evaluations rarely include both traditional econometric models and deep learning architectures within a unified evaluation framework using time-series-aware cross-validation.

Furthermore, existing studies frequently overlook the overfitting risk posed by limited quarterly macroeconomic datasets, often applying standard k-fold cross-validation that violates the temporal ordering of financial data. The use of Time Series Split cross-validation is essential to ensure that models are always trained on past data and tested.

2.3 Research Gap

This study addresses the following identified gaps in existing literature:

- A comprehensive multi-model comparison encompassing traditional econometric, classical machine learning, and modern deep learning approaches applied to Indian quarterly macroeconomic data has not been conducted.
- Most existing work does not employ time-series-aware cross-validation to robustly estimate generalization performance with limited observations.
- The use of multiple evaluation criteria (RMSE, R^2 , MAPE, directional accuracy, AUC) simultaneously to holistically assess model suitability for both regression and directional prediction tasks is underrepresented in the Indian context.
- TCN architectures have received limited evaluation in the context of Indian macroeconomic forecasting despite their strong theoretical motivations.

Chapter 3

Methodology

3.1 Theoretical Framework / Model

The study adopts a multivariate time-series regression framework in which quarterly macroeconomic indicators are used as leading predictors of India’s GDP. The core modeling hypothesis is that the joint trajectory of equity market indices, monetary policy variables, foreign capital flows, and exchange rates encodes sufficient information to forecast quarterly GDP one step ahead. The relationship is formalised as:

$$\text{GDP}(t) = f(\text{NIFTY_High}, \text{SENSEX_High}, \text{Repo_Rate}, \text{Inflation}, \text{FII_Equity}, \text{FII_Debt}, \text{FII_Total}, \text{INR_USD})$$

For deep learning models the input is a 4-step lagged sequence (one year of quarterly history) encoded as a matrix of shape 4×8 for each prediction target. For classical ML models the same sequence is flattened to a 1-D vector of length 32. Eight distinct model families spanning classical econometrics, ensemble learning, and recurrent or convolutional deep learning are evaluated within this framework to enable a comprehensive comparative analysis.

3.2 Data Collection

3.2.1 Dataset Description

The dataset was constructed by aggregating quarterly macroeconomic and financial market data for India from Q2 2004 to Q1 2025, yielding 87 quarterly observations across 11 variables. Data were sourced from four primary repositories:

- Reserve Bank of India (RBI) - Repo rate, FII equity, FII debt, FII total flows, and INR/USD exchange rate.
- Ministry of Statistics and Programme Implementation (MOSPI) - GDP at current prices (crore INR) and CPI/WPI inflation.

- Yahoo Finance - NIFTY 50 High and SENSEX High quarterly prices.

Table 3.1 provides a structural summary of all variables, their data types, missing value counts, and originating sources.

Variable	Type	Missing	Source
Quarter	datetime64	0	All sources
Nifty_High	float64	1	Yahoo Finance
Sensex_High	float64	1	Yahoo Finance
GDP (Target)	float64	1	MOSPI
Repo_Rate	float64	0	RBI
Inflation	float64	7	MOSPI / RBI
FII_Equity	int64	0	RBI
FII_Debt	int64	0	RBI
FII_Total	int64	0	RBI
INR_USD	float64	0	RBI

Table 3.1: Dataset Variable Summary (87 quarterly observations, Q2 2004 - Q1 2025)

3.3 Tools and Techniques

All analysis was implemented in Python 3. The table below lists the principal libraries used and their roles in the study.

Library / Tool	Version	Role
TensorFlow / Keras	2.19.0	LSTM, GRU, TCN model construction and training
Scikit-learn	Latest	Random Forest, XGBoost, SVM, preprocessing, CV
XGBoost	Latest	Gradient boosted trees regression
Statsmodels	Latest	ARIMA and VAR econometric models
keras-tcn	Latest	Temporal Convolutional Network (TCN) layer
Pandas / NumPy	Latest	Data wrangling and numerical operations
Matplotlib / Seaborn	Latest	Visualisation (histograms, heatmaps, loss curves)
Google Colab (GPU)	-	Computational environment for model training

Table 3.2: Software Tools and Libraries

3.4 Procedure

3.4.1 Missing Value Handling

Missing values were addressed using forward-fill interpolation (`pandas.fill()`), the standard convention for financial time series where values tend to persist across consecutive periods.

This method preserves temporal ordering and prevents information leakage from future observations. After forward-filling, zero missing values remain across all 87 observations.

3.4.2 Feature Selection and Scaling

Eight features were selected as predictor variables based on economic theory and data availability. Both features and target were independently normalised to $[0, 1]$ using scikit-learn’s `MinMaxScaler`. Separate scalers for features and target enable correct inverse transformation of model outputs for interpretability:

```
scaler_X = MinMaxScaler()
scaler_y = MinMaxScaler()
X_scaled = scaler_X.fit_transform(X) # shape: (87, 8)
y_scaled = scaler_y.fit_transform(y) # shape: (87, 1)
```

3.4.3 Correlation Analysis

A Pearson correlation analysis revealed strong positive correlations between `Nifty_High`, `Sensex_High`, `GDP`, and `INR_USD ($r = 0.91 - 1.00$), confirming co-movement of equity markets with economic output. Repo_Rate and Inflation showed negative correlations with market indices ($r = -0.31$ to -0.36), consistent with monetary contraction theory. FII_Equity exhibited modest negative correlation ($r \approx -0.20$) with indices, suggesting foreign flows are not the primary quarterly driver of index levels.`

3.4.4 Sequence Construction and Train / Test Split

Temporal sequences of length `SEQ_LEN = 4` quarters (one year of lag) were constructed using a sliding window approach, yielding 83 usable sequence-target pairs ($87 - 4 = 83$). A chronological 80/20 split produced 66 training and 17 test samples. Flattened 2D versions ($N \times 32$) were prepared for ML models requiring non-sequential input:

```
def create_sequences(X, y, seq_len):
    X_seq, y_seq = [], []
    for i in range(len(X) - seq_len):
        X_seq.append(X[i : i+seq_len]) # shape: (4, 8)
        y_seq.append(y[i + seq_len])   # scalar
    return np.array(X_seq), np.array(y_seq)

X_seq, y_seq = create_sequences(X_scaled, y_scaled, SEQ_LEN=4)
```



```
# Result → X: (83, 4, 8)    y: (83, 1)
```

```
train_size    = int(0.80 * len(X_seq))    # = 66
X_train, X_test = X_seq[:train_size], X_seq[train_size:]
y_train, y_test = y_seq[:train_size], y_seq[train_size:]
```

3.4.5 Cross-Validation Strategy

Given only 87 usable samples, a single 80/20 split may produce unstable error estimates. Five-fold `TimeSeriesSplit` cross-validation was therefore applied to the three ML models (Random Forest, XGBoost, SVM) and the two econometric models (ARIMA, VAR), ensuring the training window always precedes the validation window chronologically and preventing data leakage. Deep learning models (LSTM, GRU, TCN) were evaluated on the fixed chronological test split with `EarlyStopping` as the regularisation mechanism.

3.4.6 Model Architectures

Table 3.3 summarises the key hyperparameters and characteristics of all eight models implemented in this study. All three deep learning models share same training configura-

Model	Key Hyperparameters	Params	Notes
ARIMA	order (2,1,2)	—	Univariate; raw GDP scale
VAR	maxlags = 2, ic = AIC	—	Multivariate; 1 fold skipped
Random Forest	n_estimators = 100, max_depth = 5	—	Flattened 2D input ($N \times 32$)
XGBoost	n_estimators = 100, learning_rate = 0.05, max_depth = 3, subsample = 0.8	—	L1/L2 regularised
SVM	RBF kernel, $C = 1.0$, $\epsilon = 0.1$	—	Normalised 2D input
LSTM	32 units, dropout = 0.2, recurrent dropout = 0.1	5,281	Stopped at epoch 63
GRU	32 units, dropout = 0.2, recurrent dropout = 0.1	4,065	Stopped at epoch 128
TCN	32 filters, kernel size = 2, dilations = [1,2,4]	11,265	Stopped at epoch 188

Table 3.3: Model Hyperparameter Summary

tion: Adam optimiser, MSE loss, batch size of 16, up to 300 epochs, a validation split of 20%, and `EarlyStopping` with patience = 30 while restoring the best weights. The GRU architecture is illustrated below:

```

model_gru = Sequential([
    Input(shape=(4, 8)),
    GRU(32, dropout=0.2, recurrent_dropout=0.1),
    Dense(1)
])
model_gru.compile(optimizer='adam', loss='mse')
# Total trainable parameters: 4,065

```

3.4.7 Evaluation Metrics

Model performance was assessed using five complementary metrics to ensure a complete picture of predictive utility:

- RMSE (Root Mean Squared Error) - Average prediction error magnitude; sensitive to large outliers; reported in original GDP scale (crore INR) for interpretability.
- R^2 (Coefficient of Determination) — Proportion of GDP variance explained; ranges from $-\infty$ to 1.0; negative values indicate worse performance than a simple mean predictor.
- MAPE (Mean Absolute Percentage Error) - Scale-independent percentage error; enables comparison across different GDP magnitude periods.
- Directional Accuracy - Fraction of quarters where the model correctly predicts whether GDP will increase or decrease (binary 50% = random baseline).
- AUC (Area Under the ROC Curve) - Discriminative power for directional prediction; 0.5 = random chance, 1.0 = perfect.

Chapter 4

Results and Analysis

4.1 Results

4.1.1 Regression Performance

Table 4.1.1 presents the complete regression performance of all eight models on the held-out test set (17 quarterly observations). RMSE is expressed in the original GDP scale (crore INR). Models are ranked by RMSE (ascending).

Rank	Model	RMSE (Cr INR)	R^2	MAPE %
1st	GRU	22,55,120	0.6597	4.69%
2nd	LSTM	27,41,261	0.4972	5.35%
3rd	ARIMA	58,79,892	-1.3131	11.52%
4th	XGBoost	97,70,651	-5.3872	19.49%
5th	Random Forest	1,00,74,636	-5.7908	20.16%
6th	SVM	1,38,81,144	-11.8918	28.90%
7th	TCN	1,66,36,273	-17.5171	28.96%

4.1.2 Directional Accuracy and Classification Performance

Table 4.1.2 presents the directional accuracy and AUC of the six models for which classification evaluation was conducted. Directional accuracy measures whether the model correctly predicts the sign of GDP change; 50% represents the random baseline.

Model	Directional Accuracy	AUC
GRU	75.00%	0.5833
LSTM	68.75%	0.5417
TCN	62.50%	0.5000
Random Forest	50.00%	0.5000
SVM	50.00%	0.5833
XGBoost	37.50%	0.4167

4.1.3 Cross-Validation Results

Table 4.1.3 reports five-fold TimeSeriesSplit cross-validation results. CV RMSE for ML models is in the normalised $[0, 1]$ scale; econometric models are in the original GDP scale.

Model	CV Mean RMSE	CV Std RMSE
Random Forest	0.1404	0.0340
XGBoost	0.1409	0.0361
SVM	0.2354	0.0699
VAR	29,77,104	18,81,496
ARIMA	44,02,955	14,78,195

4.1.4 Best Model Summary

Table 4.1.4 consolidates the best-performing model across each evaluation criterion.

Criterion	Best Model	Value
Lowest Test RMSE	GRU	22,55,120 Cr INR
Highest R^2	GRU	0.6597
Lowest MAPE	GRU	4.69%
Best Directional Accuracy	GRU	75.00%
Best CV RMSE (normalised)	Random Forest	0.1404

4.2 Analysis

4.2.1 Deep Learning vs. Classical Approaches

GRU and LSTM decisively outperform all classical and ML models on the test set. GRU’s RMSE of approximately ₹22.6 lakh crore is 18% lower than LSTM and more than four times lower than ARIMA (₹58.8 lakh crore). GRU’s R^2 of 0.6597 indicates that roughly 66% of the variance in quarterly GDP is explained, a substantial improvement over LSTM (49.7%). All other models - XGBoost, Random Forest, SVM, and TCN - exhibit negative R^2 values, meaning their predictions are worse than simply predicting the training mean for every quarter.

The MAPE of 4.69% for GRU is particularly noteworthy: quarterly GDP predictions deviate by less than 5% from the true value on average, a practically meaningful accuracy level for macroeconomic forecasting. LSTM’s MAPE of 5.35% is also competitive. In contrast, tree-based models (XGBoost: 19.49%, Random Forest: 20.16%) and SVM (28.90%) exhibit substantially higher percentage errors on the original scale.

4.2.2 Directional Prediction

GRU correctly predicts the direction of quarterly GDP change 75% of the time across 16 directional test periods, substantially above the 50% random baseline. LSTM achieves 68.75%. Notably, TCN - despite poor regression performance ($R^2 = -17.52$) - achieves 62.5% directional accuracy, suggesting that sequential convolutional features retain some directional information even when absolute magnitude estimates fail. XGBoost’s below-chance directional accuracy (37.5%) confirms systematic directional errors, consistent with its negative R^2 on the test set.

4.2.3 ML vs. Deep Learning: The CV Paradox

Random Forest and XGBoost achieve the two lowest cross-validation RMSE values (0.1404 and 0.1409 on the normalised scale), apparently outperforming all other models. Yet on the held-out test set, both models produce negative R^2 and MAPE values exceeding 19%. This divergence arises from a scale mismatch: CV RMSE is computed on MinMax-normalised targets, compressing GDP variability in early training folds (2004–2015). When predictions are inverse-transformed to the original scale, the errors magnify substantially, exposing overfitting to the normalised training dynamics rather than true GDP growth patterns. This finding underscores that normalised CV RMSE alone is insufficient for evaluating macroeconomic forecasting models; evaluation in the original scale using R^2 , MAPE, and directional accuracy is essential.

4.2.4 Deep Learning Training Dynamics

All three deep learning models employed EarlyStopping (patience = 30) with restoration of best weights. Their convergence trajectories differed substantially:

- LSTM — Converged at epoch 63 with stable validation loss. The lightweight 5,281-parameter architecture provided adequate representational capacity for the 4-step quarterly sequences.
- GRU — Converged at epoch 128, exhibiting slower but more progressive validation loss reduction. GRU’s simplified gating (4,065 parameters) allowed richer temporal pattern extraction with less overfitting risk than LSTM.
- TCN — Required 188 epochs before early stopping, with erratic validation loss throughout. The dilated convolution receptive field (14 steps with dilations [1, 2, 4] and kernel size 2) substantially exceeds the available sequence length of 4 quarters, preventing effective use of dilations and causing instability.

Representative GRU training log:

```
Epoch   1/300 | loss: 0.1012  val_loss: 0.1453
Epoch  20/300 | loss: 0.0086  val_loss: 0.0096
Epoch  98/300 | loss: 0.0092  val_loss: 0.0058  ← best weights
Epoch 128/300 | loss: 0.0069  val_loss: 0.0077  ← EarlyStopping
```

GRU → Test RMSE: 0.0569 R^2 : 0.6597 Stopped @ epoch: 128

4.3 Validation

4.3.1 Cross-Validation Robustness

Five-fold TimeSeriesSplit cross-validation was applied to all classical and ML models to guard against instability from the small sample size. Random Forest’s CV RMSE of 0.1404 ± 0.0340 and XGBoost’s 0.1409 ± 0.0361 indicate consistent in-distribution performance across folds. SVM’s higher standard deviation (0.0699) suggests greater sensitivity to the distribution of earlier folds. The VAR model encountered a degrees-of-freedom constraint in one fold (the smallest training window lacked sufficient observations for the 9-variable system with 2 lags), and that fold was skipped; the reported CV RMSE is based on the remaining four folds.

4.3.2 EarlyStopping as Regularisation for Deep Learning

For deep learning models, the validation split (20% of training data, i.e., approximately 13 samples) served as the held-out regularisation signal. EarlyStopping with patience = 30 and best-weight restoration ensured that the reported test performance reflects the checkpoint with lowest validation loss rather than the final epoch, reducing the risk of overfitting to the small training set. The gap between training loss and validation loss remained small for both LSTM (train MSE ≈ 0.005 , val MSE ≈ 0.006) and GRU (train MSE ≈ 0.007 , val MSE ≈ 0.006), confirming that neither model severely overfit the training data.

4.3.3 Inverse-Transform Verification

A critical validation step was verifying that all reported metrics (RMSE in crore INR, R^2 and MAPE) were computed after inverse-transforming normalised predictions back to the original GDP scale using the fitted `scaler.y`. This ensures that the reported absolute error values are interpretable in real economic terms and that comparisons between models operating at different scales (econometric models at raw scale vs. ML/DL models at normalised scale) are made consistently on the same original scale.

```

# Inverse transform predictions to original GDP scale
y_test_inv = scaler_y.inverse_transform(y_test)
pred_gru_inv = scaler_y.inverse_transform(pred_gru)

# Compute metrics on original scale
rmse = np.sqrt(mean_squared_error(y_test_inv, pred_gru_inv))
r2 = r2_score(y_test_inv, pred_gru_inv)
mape = np.mean(np.abs((y_test_inv - pred_gru_inv) / y_test_inv)) * 100

# GRU result: RMSE = 2255120  R2 = 0.6597  MAPE = 4.69%

```

4.3.4 Limitations of the Validation Framework

Several methodological limitations of the current validation framework should be acknowledged:

- Deep learning models were not subject to TimeSeriesSplit CV due to computational constraints; their test-set performance is based on a single chronological split (17 samples), which may not fully reflect performance across different market regimes.
- The test period (approximately 2021–2025) coincides with the post-COVID economic recovery phase, which may be structurally different from the training period, potentially inflating errors for models that learned pre-COVID patterns.
- With only 17 test samples, directional accuracy estimates carry high variance; a 75% accuracy on 16 directional comparisons (GRU) corresponds to 12 correct predictions, and the confidence interval for the true accuracy is wide.
- The normalized scale used for ML model CV RMSE and the original scale used for test set evaluation are not directly comparable, and combining both in a single ranking table risks misleading interpretation if not contextualised carefully (as discussed in Section 4.2.3).

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This study presents a comprehensive investigation into the relationship between macroeconomic indicators and Indian stock market performance, with a specific focus on forecasting quarterly GDP using a diverse portfolio of modeling approaches. The key findings are as follows

- Deep learning models significantly outperform traditional econometric approaches (ARIMA, VAR) and classical machine learning models (Random Forest, XGBoost, SVM) in both regression accuracy and directional prediction for Indian macroeconomic time series.
- The GRU (Gated Recurrent Unit) model is the overall best performer, achieving the lowest RMSE (approximately 22.6 lakh crore INR), highest R^2 (0.6597), lowest MAPE (4.69%) and highest directional accuracy (75.00%) on the test set. These results confirm the hypothesis that recurrent architectures with efficient gating mechanisms are particularly well-suited for modeling quarterly macroeconomic dynamics.
- LSTM achieves competitive performance ($R^2 = 0.4972$, directional accuracy 68.75%), validating the role of long-range memory mechanisms in capturing macroeconomic temporal dependencies.
- TCN, despite theoretical motivations, underperforms on this dataset due to the mismatch between the short sequence length (4 quarters) and its dilated receptive field architecture.
- Tree-based ensemble methods (Random Forest, XGBoost) exhibit a misleading pattern of low normalized CV RMSE but poor out-of-sample generalization in the

original GDP scale, highlighting the importance of evaluating financial forecasting models in the original economic scale using multiple criteria.

- Macroeconomic variables, particularly GDP, INR/USD exchange rate, NIFTY/SENSEX market levels, repo rate, and FII flows, exhibit significant and economically interpretable inter-correlations consistent with monetary and capital markets theory.
- Overall, the study confirms that macroeconomic indicators significantly influence Indian stock market dynamics, and that deep learning architectures, particularly GRU, provide a powerful and practically viable framework for forecasting macroeconomic trends and market movements in India.

5.2 Limitations

This study acknowledges the following limitations:

- Dataset size: The quarterly dataset comprises only 87 observations (83 usable after sequence creation). This severely constrains the capacity of deep learning models, which typically require thousands of observations to fully realize their potential. Results should be interpreted with appropriate caution regarding generalizability.
- Exclusion of external shocks: Geopolitical events, global financial crises (e.g., the 2008 Global Financial Crisis, COVID - 19 pandemic), and global commodity price shocks are not explicitly modeled. These events introduce structural breaks that significantly distort macroeconomic-financial relationships.
- No investor sentiment data: Behavioral and sentiment factors, which are known to drive short-term market deviations from fundamentals, were not included due to data availability constraints.
- TCN hyperparameter sensitivity: The TCN architecture requires more careful hyperparameter tuning (particularly dilation factors and kernel size) relative to sequence length, which was not fully explored in this study.
- Model interpretability: While GRU achieves the best predictive performance, its internal representations are not directly interpretable in economic terms, limiting actionability for policymakers.

5.3 Future Scope

The following directions are identified for future research:

- High-frequency and real-time data: Extending the analysis to monthly or weekly macroeconomic indicators and incorporating real - time RBI/MOSPI data streams would increase the number of observations, potentially enabling more complex architectures.
- Sentiment analysis integration: Incorporating NLP - derived sentiment scores from financial news, RBI monetary policy statements, and social media could capture the behavioral dimension of market movements.
- Hybrid architectures: Combining GRU/ LSTM with attention mechanisms or with econometric structural equation models (VAR-LSTM hybrid) may further improve forecasting accuracy.
- Extended macroeconomic feature set: Incorporating additional variables such as crude oil prices, global equity market indices (S&P 500, Hang Seng), credit default swap spreads, and manufacturing PMI could enrich the feature space.
- Probabilistic forecasting: Implementing Bayesian LSTM or Monte Carlo Dropout approaches to generate prediction intervals and quantify forecast uncertainty would significantly improve practical utility for risk management applications.
- Explainability: Applying SHAP (SHapley Additive exPlanations) or attention-based interpretability methods to the GRU model would provide economically interpretable attribution of predictions to specific macroeconomic drivers.

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